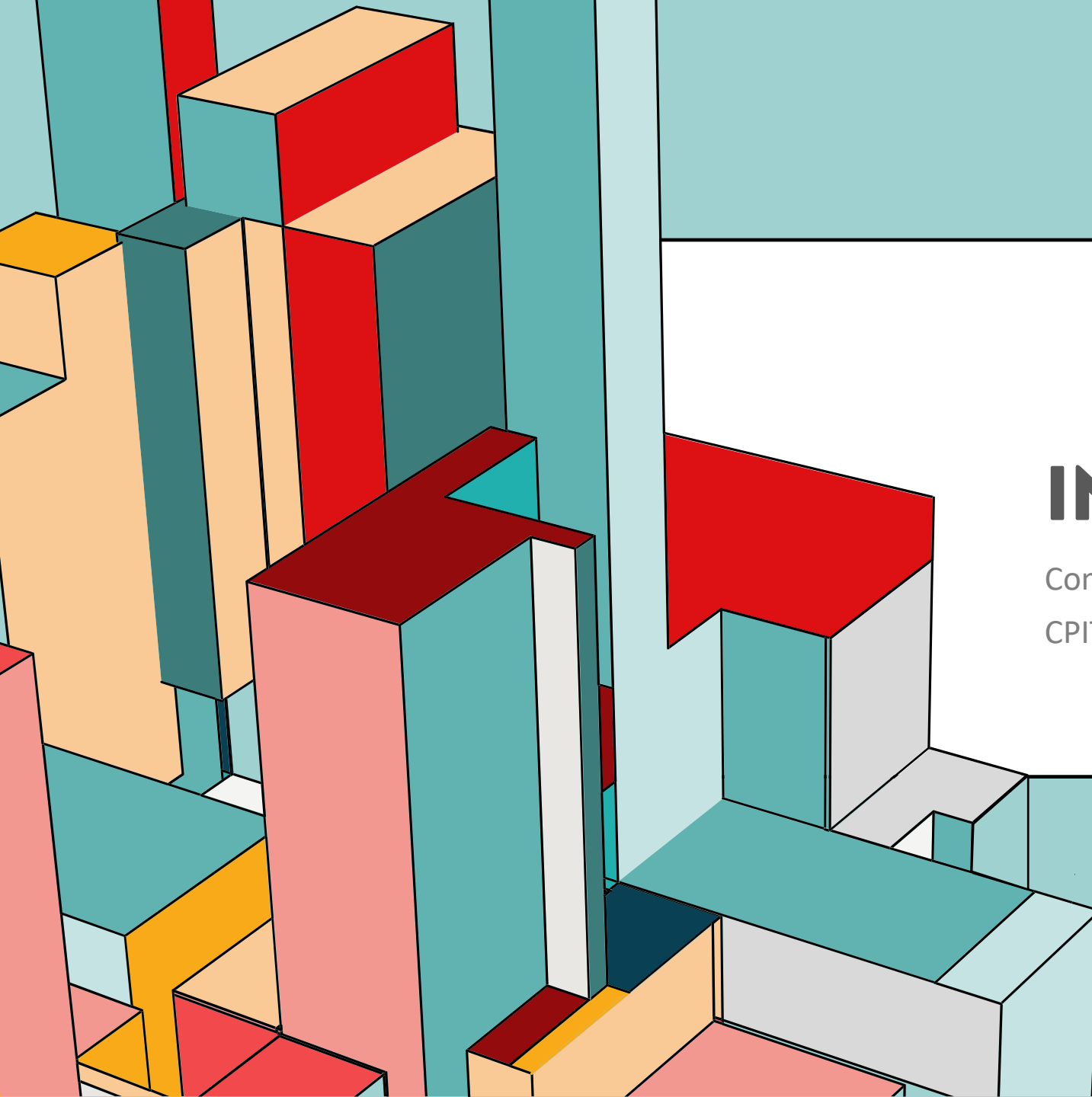


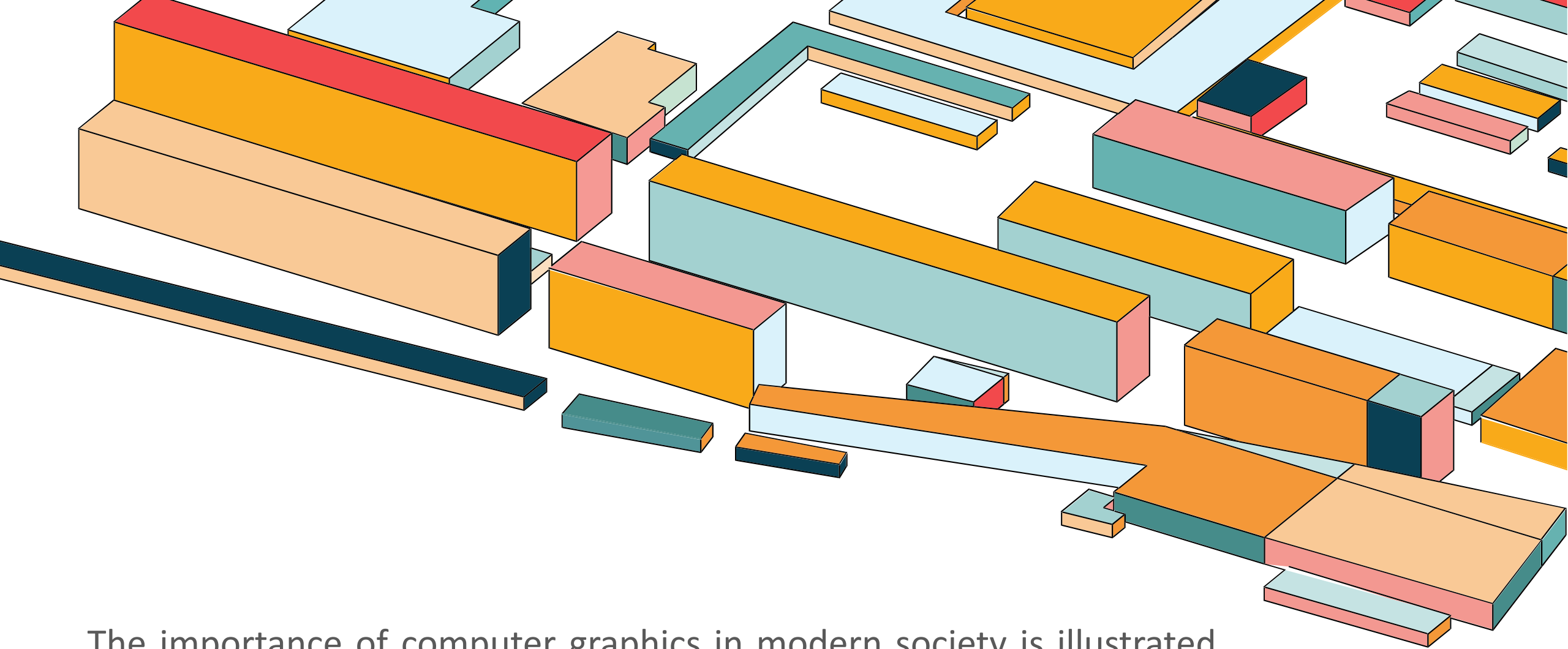
An abstract composition of various 3D rectangular blocks and prisms. The blocks are rendered in a variety of colors including teal, orange, red, pink, and light blue. They are arranged in a way that creates a sense of depth and perspective, with some blocks appearing to be stacked or connected. The blocks have black outlines, giving them a clean, graphic appearance. The background is a solid light blue color.

INTRODUCTION

Computer Graphics

CPIT285

Dr. Rayed ALGhamdi



The importance of computer graphics in modern society is illustrated by the great quantity and variety of applications and their impact on our daily lives. Computer graphics can be two-dimensional (2D) or three-dimensional (3D), animated, and interactive...

WHAT IS COMPUTER GRAPHICS?

- Images produced by computer
- What can be classified a graphics?
 - Photoshop images – Graphs - Fonts
 - Maps – 3D games – User Interfaces
 - Buildings Designs – Web Pages
- Can ... be called graphics?
 - Videos
 - Animations
 - Computer Games



WHAT IS COMPUTER GRAPHICS?

- The study of creating, manipulating, and using visual images on the computer.
- **Graphics deal with all aspects of processing images using a computer...**
 - Hardware
 - Software
 - Applications



COMPUTER GRAPHICS APPLICATIONS

- Movies/films/cinema
- Computer/video games
- Simulation
- CAD (Computer-Aided Design)
- Visualization
- Virtual/Augmented reality
- Graphical User Interfaces
- Images processing
- Education & Training
- Computer Art



COMPUTER GRAPHICS APPLICATIONS

Entertainment Applications

- Graphics extensively used in producing today's movies & T.V shows
 - Real actors with graphics effects
 - Full Graphical Acting



Chroma Screen/ keyer



Motion Capture



مسلسلات آل رشي



Gollum
Lord of the rings

COMPUTER GRAPHICS APPLICATIONS

Entertainment Applications

- 2D & 3D Animations



COMPUTER GRAPHICS APPLICATIONS

Entertainment Applications

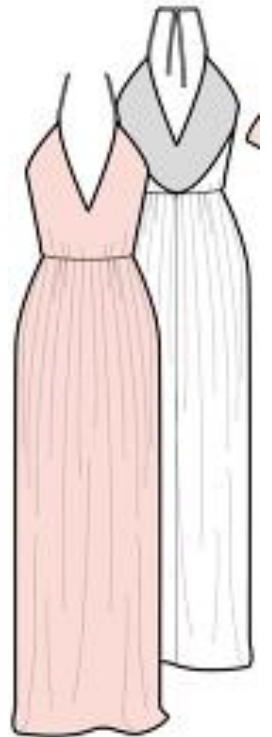
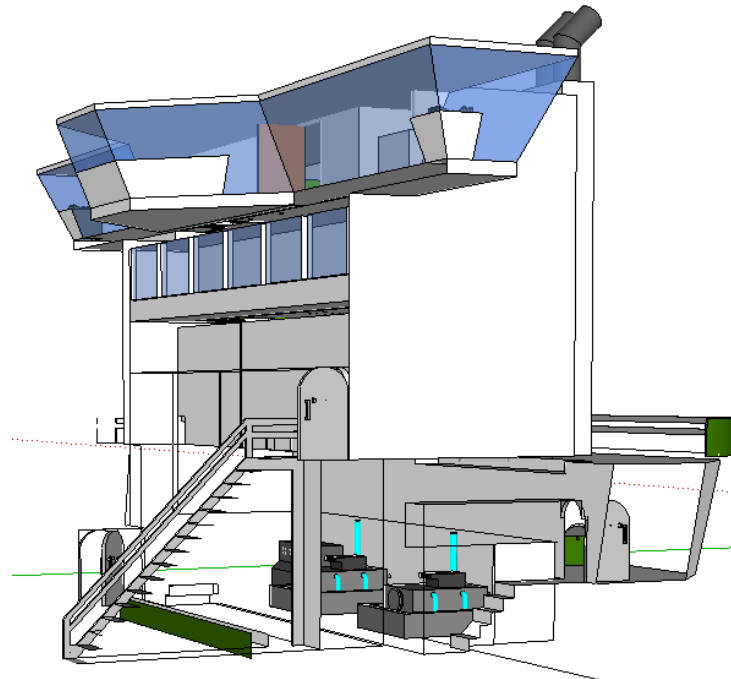
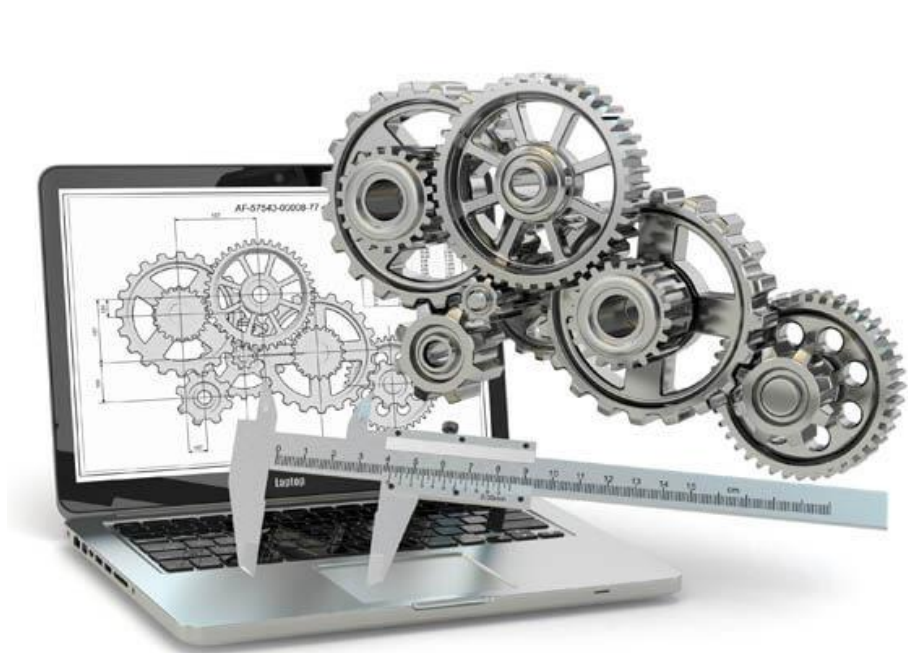
- The core element in computer gaming



COMPUTER GRAPHICS APPLICATIONS

Designs & Engineering Applications

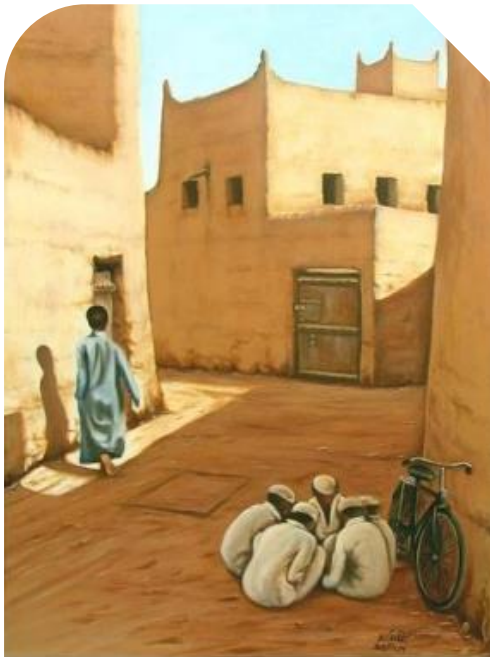
- Computer Aided Design (CAD) is routinely used in the design of buildings, cars, airplanes, computers, textiles, and many other products.



COMPUTER GRAPHICS APPLICATIONS

Art Applications

- Graphics techniques are widely used in art applications.
- Artists use a variety of computer applications for processing & publishing ...



COMPUTER GRAPHICS APPLICATIONS

Simulating Applications

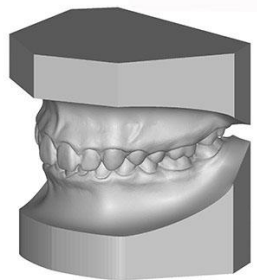
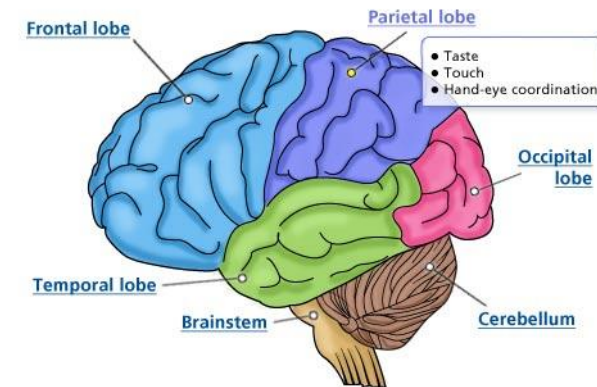
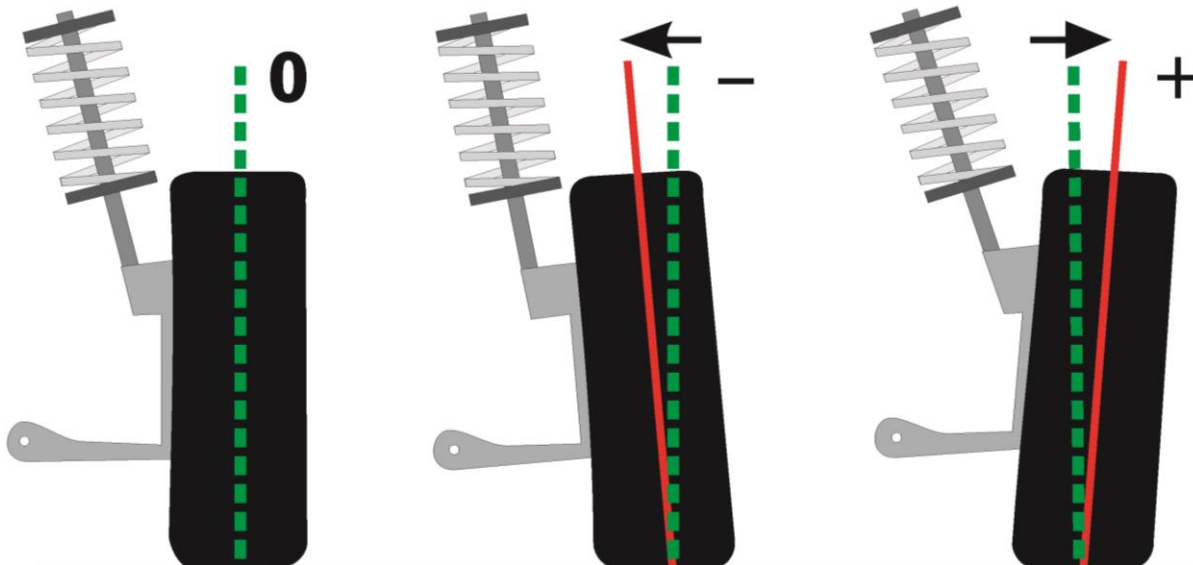
- Simulation is an essential component of modeling in many scientific applications. It helps to provide virtual experience before the real practicing; resulting in time, effort and money-saving.



COMPUTER GRAPHICS APPLICATIONS

Visualizing Applications

- Visualization is used in many fields, such as medicine, engineering, education, physics and biology ... They use computers to make visual models to research and diagnose the situation under study.



COMPUTER GRAPHICS APPLICATIONS

Virtual & Augmented Realities Applications

- **Augmented Reality** applies unreal objects to a real environment, in which a view of reality is modified by a computer.
- **Virtual Reality** uses physical objects in a virtual or imaginary environment.

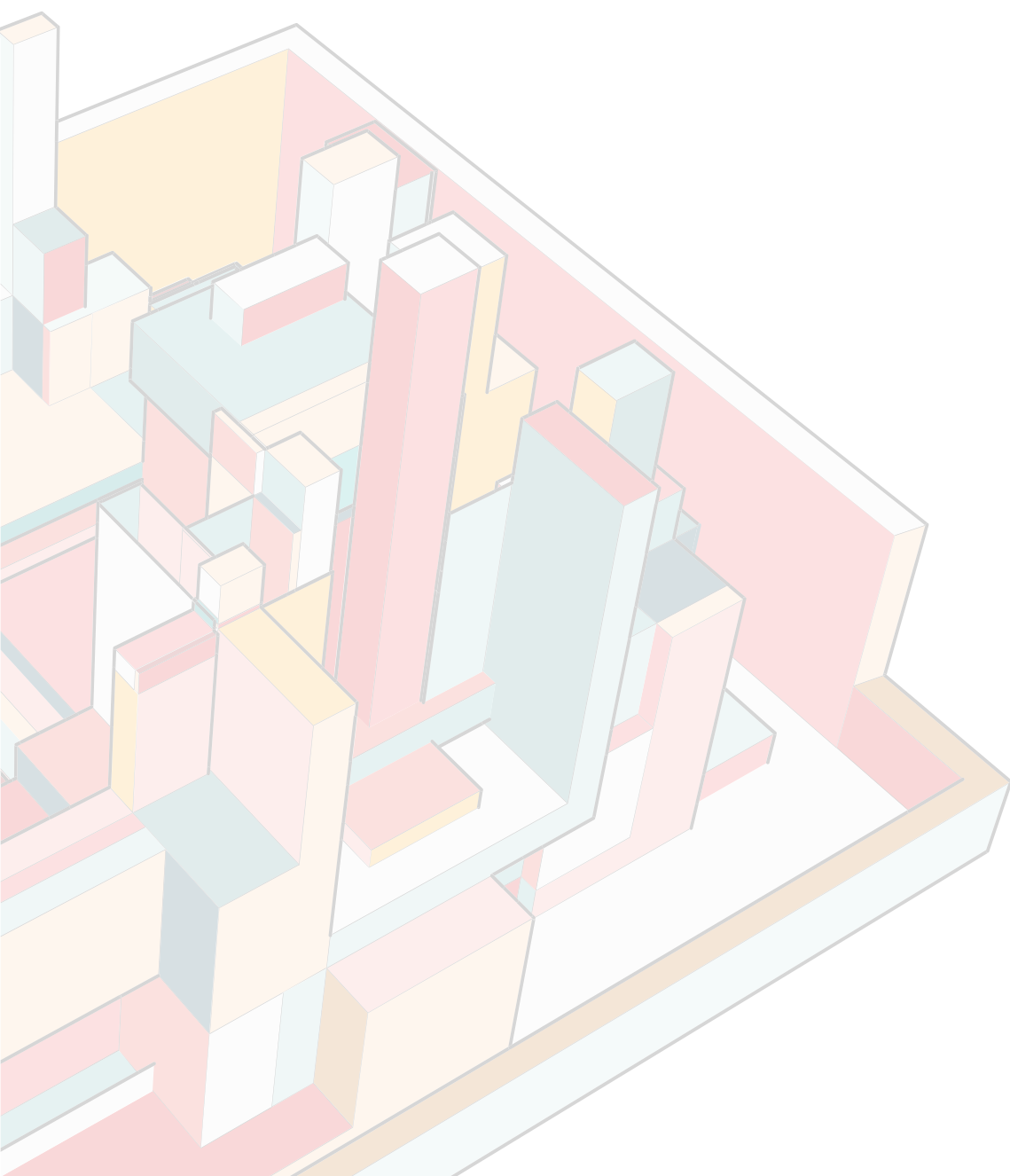


COMPUTER GRAPHICS APPLICATIONS

Interfaces Applications

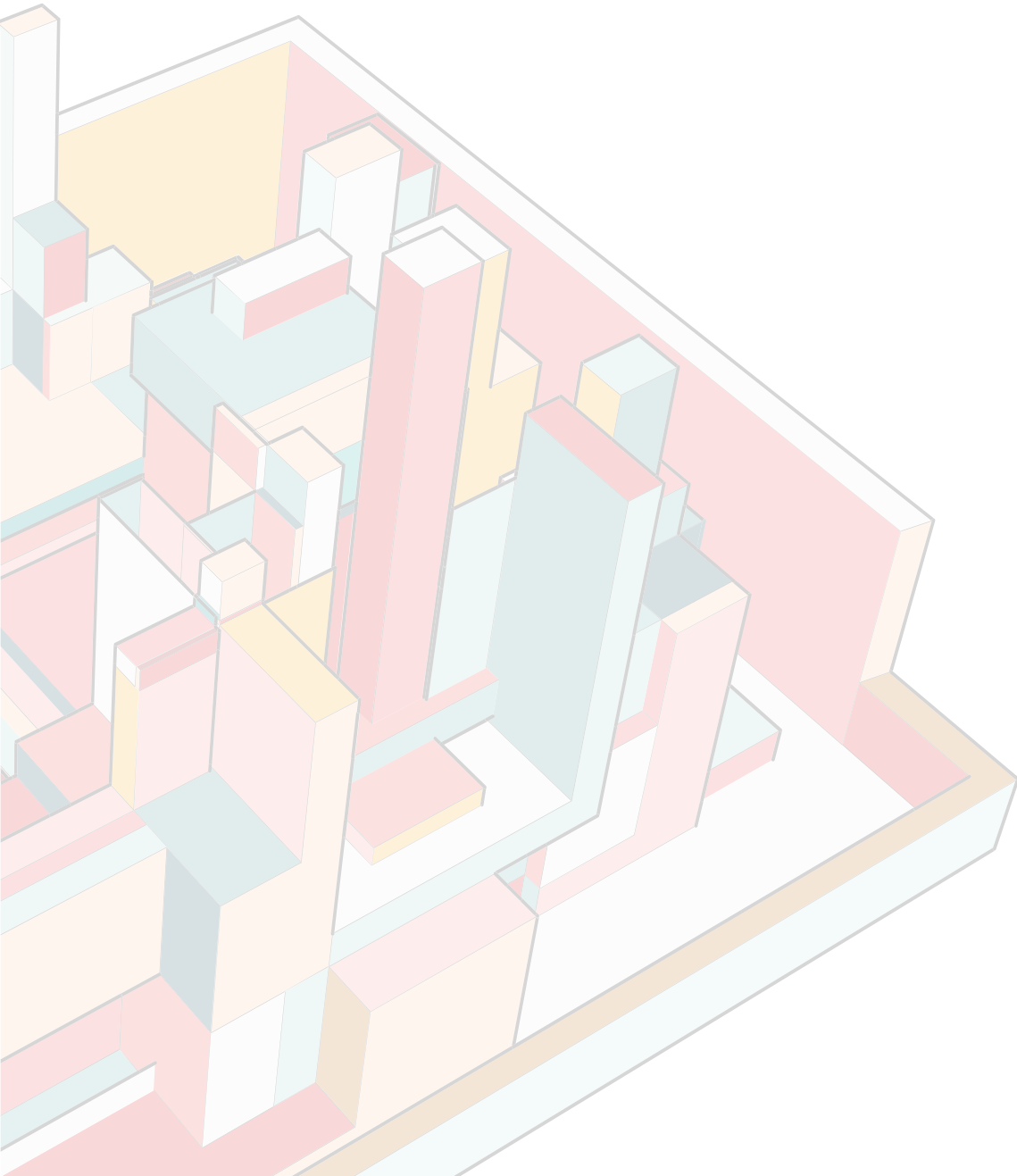
- A user interface for a system, app, website... is a key success factor based on graphical uses to facilitate the use of the system...





WHY DO YOU STUDY THIS COURSE?

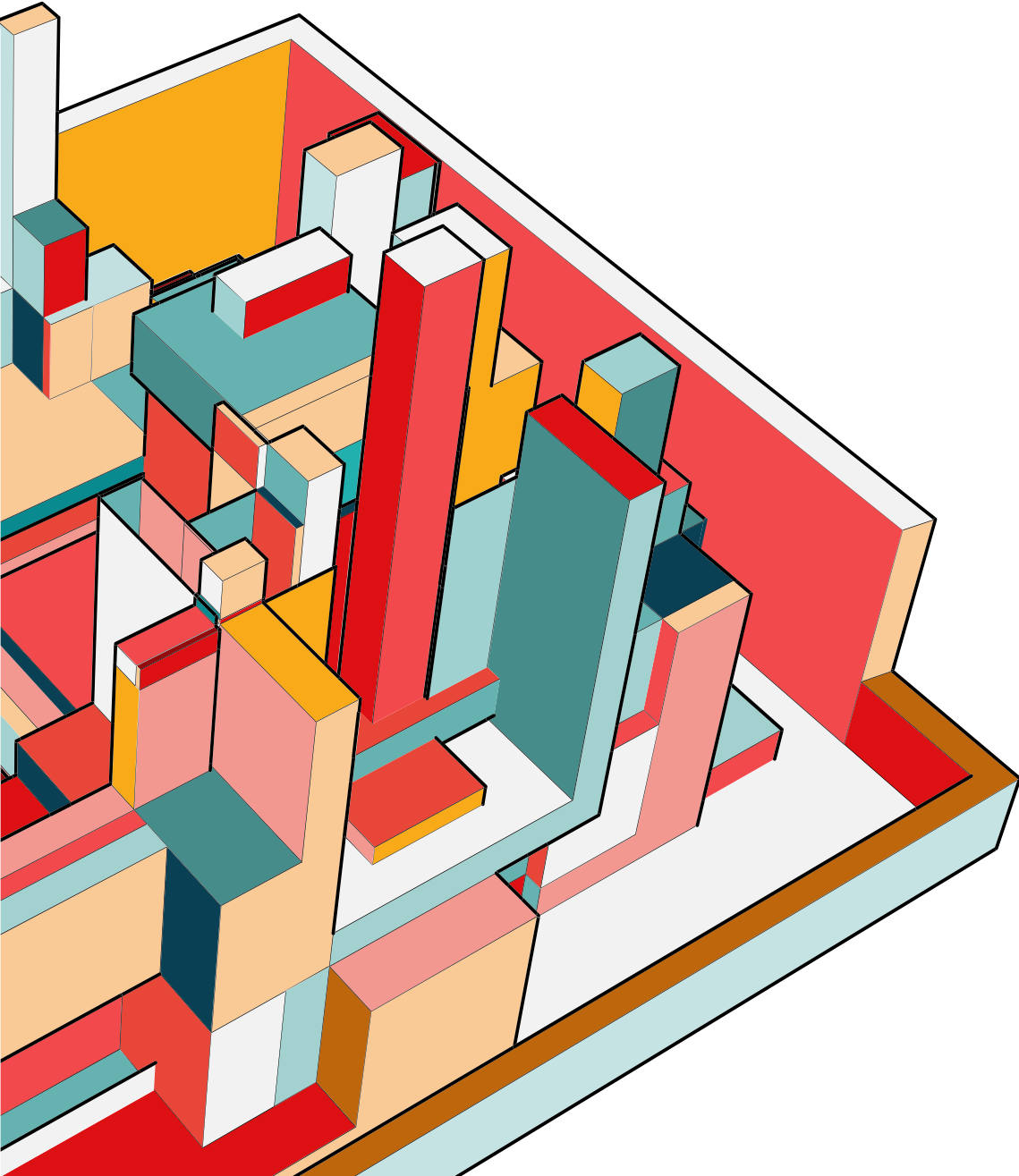




WHAT DO YOU EXPECT TO LEARN?



..... your answer to the previous question determines your expectations.



CPIT285 SYLLABUS

- **Description**
- **Covered Topics**
- **Course Learning Outcomes (CLOs)**
- **Assessments**
- **Roles of Lectures and Labs**

COURSE DESCRIPTION

- This course introduces the basic concepts of computer graphics. It provides the necessary theoretical background and demonstrates the application of computer science to graphics.
- The course covers fundamental topics of 2D and 3D graphics. It includes 2D/3D graphics representations, transformations, the viewing pipeline, visibility, lighting, textures, animation, and rendering.

COVERED TOPICS

- Graphic Hardware/Software Systems
- Computer Graphics APIs
- Introduction to mathematical process (linear algebra) of the graphics
- Graphics primitives
- 2D/3D Graphics Pipelines
- 2D/3D Modelling Transformations
- 2D/3D Viewing Transformations
- Lighting and Shading
- UV mapping & Texturing
- Animation
- Rendering

COURSE LEARNING OUTCOMES (CLOS)

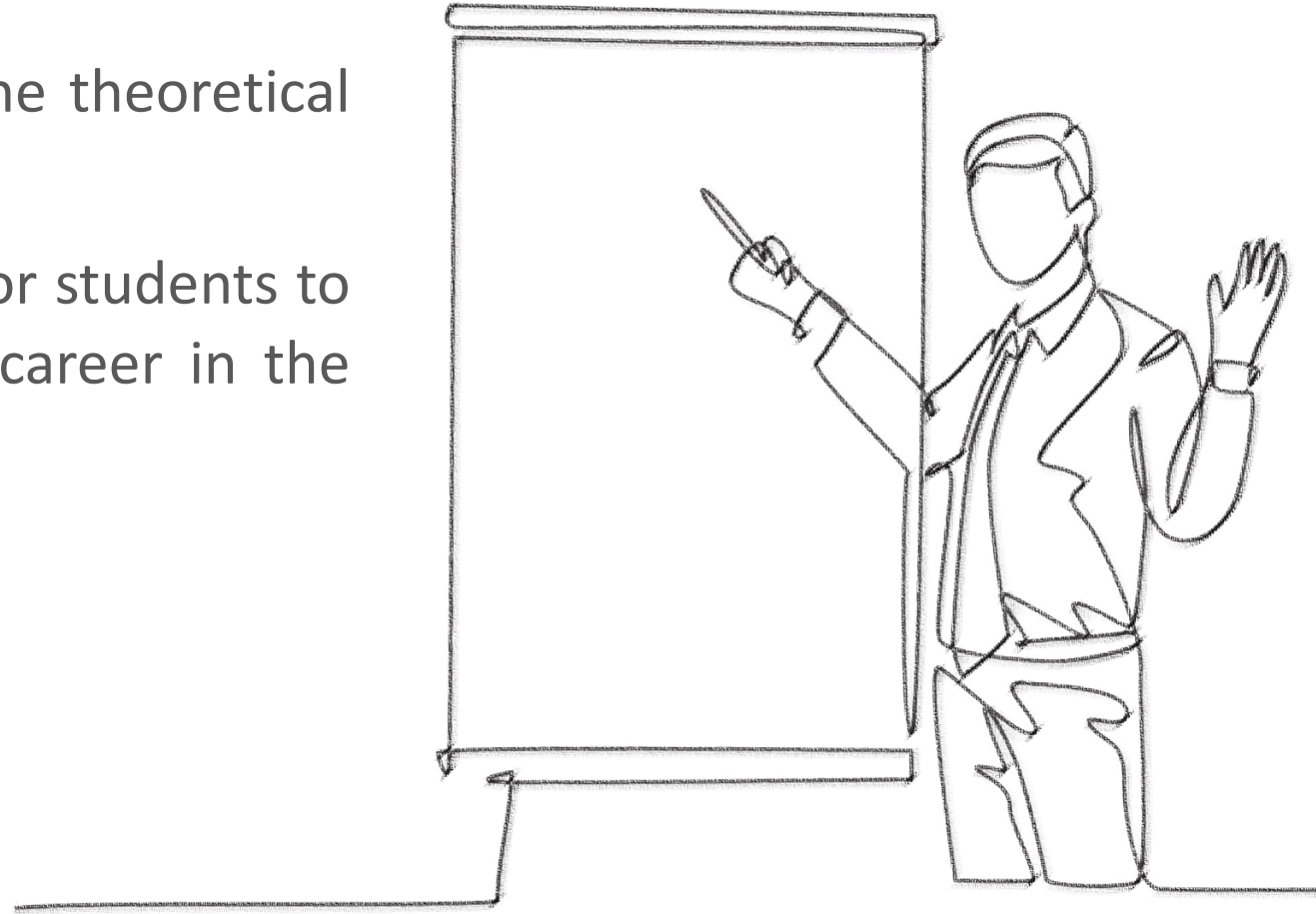
1. Understand the graphics hardware and software systems
2. Understand the core concepts & mathematical foundations of computer graphics
3. Realize the role of graphics APIs
4. Deploy the coloring harmony schemes
5. Identify the structure of 2D/3D graphics
6. Modeling 3D objects
7. Layout good UVs (two-dimensional texture coordinates) of their models
8. Create textures for 3D models
9. Apply lighting effects for 3D models
10. Set up cameras for 3D models
11. Animate 2D/3D objects
12. Create and present the final graphics renders
13. Design and implement a complete graphical project in groups

ASSESSMENTS

- A student progress is assessed as follows:
 1. Activity 1: Model your personal head (5%)
 2. Activity 2: Bring your head model into life (5%)
 3. Lab Work: ongoing interaction and work submitted on Discord (20%)
 4. Group Project: A project that implements all the core concepts in this course (20%)
 5. Midterm Exam (20%)
 6. Final Exam (30%)

ROLES OF LECTURES AND LABS

- The role of lecturing is to explain the theoretical concepts of 2D/3D graphics.
- Lecturing plays a motivational role for students to set up personal goals for a future career in the graphics/gaming industry.



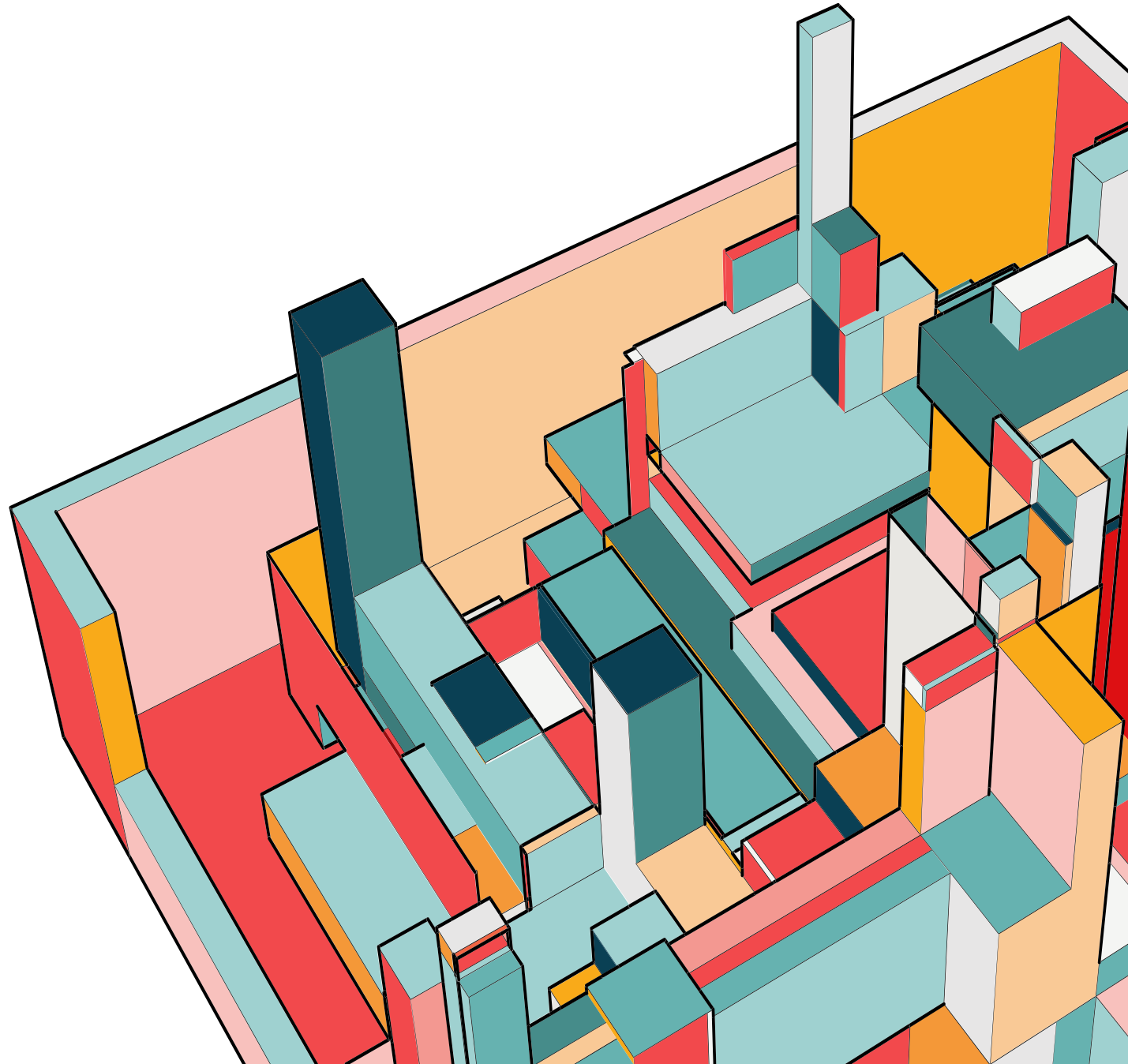
ROLES OF LECTURES AND LABS

- Since the students are great users of Discord, Discord is used as the primary communication platform for lab mentoring and encouraging knowledge/experience sharing, and peer learning. Please [JOIN CPIT285 Discord Server](#). Students Labs work differently from normal classes. The students gather in a lab channel in Discord where everyone can share their screen at the same time.
- A teaching assistant is always present to help answer any questions or troubleshoot problems, but the main purpose of the student lab is for the students to help each other. The instructor is there in the labs to help lead the group and guide the flow of the lab. As the course progresses, the instructor takes less of a role in the labs to gradually release responsibility to the students.
- The Student Labs are there to help build community and teamwork that lasts long after they complete the course.

100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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